# > 2026 MIDWEST BIOINFORMATICS TRAINING

**TUESDAY & WEDNESDAY, MAY 12-13, 2026**

MDHHS Bureau of Laboratories
Lansing, Michigan

# > MIDWEST BIOINFORMATICS TRAINING

AGENDA OVERVIEW

## DAY 1

*Tuesday, May 12, 8:30am-4:30pm*

### > BIOINFORMATICS IN PUBLIC HEALTH

- Welcome & introductions
- Conceptual overview: Genomic data and bioinformatics for public health
- Technical overview: Pipelines, workflows, and programs, oh my!

### > TRAINING MODULE 1: GETTING TO KNOW THE PIPELINE

- Introduction to GiHub Codespaces
- Run sequencing reads from a single bacterial sample through each step of a basic genomic analysis pipeline including quality control, read trimming, and de novo assembly

### > TRAINING MODULE 2: END-TO-END PIPELINE WITH REFERENCE-BASED ASSEMBLY

- Introduction to reference-based assembly
- Run sequencing reads from a single bacterial sample through the pipeline in end-to-end fashion with reference-based genome assembly

## DAY 2

*Wednesday, May 13, 8:30am-4:00pm*

### > TRAINING MODULE 3: CLUSTER ANALYSIS OF BACTERIAL GENOMES

- Day 2 overview & setup
- Introduction to phylogenetic analysis
- Generate a phylogeny and SNP matrices from a sample set of assembled bacterial genomes
- Visualize outputs for comparison of samples in a cluster

### > USEFUL APPLICATIONS & BEST PRACTICES

- Common errors & troubleshooting panel discussion
- Useful tools & resources
- Best practices & downstream applications
- Wrap up & conclusions